



COLLEGE
OF
MEDICINE

Cholera

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BACKGROUND

- Cholera is an acute diarrheal infection caused by ingestion of water or food contaminated with the bacterium *Vibrio cholerae*.
- Researchers have estimated that each year there are **1.3 million to 4.0 million cases** of cholera, and **21 000 to 143 000 deaths** worldwide due to cholera. (WHO Jan. 2019)



Characteristic of the Bacteria

□ *Vibrio cholerae*, discovered in 1883 by Robert Koch during a diarrheal outbreak in Egypt.

□ *V. cholerae* has 2 major biotypes: classical and El-Tor

□ The organism is a comma-shaped, gram-negative, aerobic bacillus, size varies from **1-3 mm in length to 0.5-0.8 mm in diameter.**

seen on light microscopy

Vibrio cholera



- Its antigenic structure consists of a flagella H antigen and a somatic O antigen.
- It is the differentiation of the latter that allows for separation into pathogenic and nonpathogenic strains (O1 and O139 are pathogenic sero-groups)

determined in Baghdad (Iraq)



□ The O1 serogroup is divided into the main **serotypes Inaba and Ogawa**, and O1 is subdivided into two distinct biotypes, designated classical and El Tor



Epidemiology

- Since 1817, there have been 7 cholera pandemics.
- The first 6 occurred from 1817-1923 and were caused by *V. cholerae*, the classical biotype.
- The pandemics originated in Asia with subsequent spread to other continents.



- The seventh pandemic began in Indonesia in 1961 and affected more countries and continents than the previous 6 pandemics.
- It was caused by *V. cholerae* El-Tor.



Epidemiology (continued)

- Crowding & gathering of people during religious rituals enhance the spread of infection
- Index cases when travelled back to their homes may pass the organism to at risk individuals leading to secondary epidemic or small scale infection

PATHOGENESIS

- *V cholerae* cause clinical disease by producing an enterotoxin (exotoxin) ^{→ which cause manifestation not the bacteria like diphtheria} that **promotes the secretion of fluid and electrolytes into the lumen of the gut** ^{→ normally they were reabsorbed} (through its effect on adenylate cyclase- Cyclic AMP system of mucosal cells of small intestine)



PATHOGENESIS

- The result is watery diarrhea with electrolyte concentrations isotonic to those of plasma. *leading to dehydration*
- The enterotoxin acts locally & does not invade the intestinal wall. As a result few WBC & no RBC are found in the stool

**death may occur by hypovolemic shock or renal shutdown*

- The average cholera CFR reported globally in 2021 was 1.9% (2.9% in Africa), **a significant increase above acceptable (<1%)** and the highest recorded in over a decade. *→ Case-fatality rate*
- The number of reported cholera cases increased by 13% and deaths by 71% in 2023 compared to 2022. Over 4000 people died last year (2023) from a disease that is preventable and easily treatable.

TRANSMISSION

- Cholera is transmitted by the fecal-oral route through contaminated water & food. Person to person infection is rare.
by person to
lower cases localized to people ate from same restaurant
more cases and strains
- The infectious dose of bacteria required to cause clinical disease varies with the source. **Infective dose in normal human is estimated to be 10^{11} organism** to produce a clinical disease
lower dose in people that have taken anti-acid → since stomach acidity kills



AT RISK GROUPS

- All ages but children & elderly are more severely affected.
- Subjects with blood group “O” are more susceptible; the cause is unknown.
- Subjects with **reduced gastric acid**



Incubation and infectious period

- **Incubation period:** it is few hours to 5 days but typically 2-3 days.
- **Infectious period:** During the acute stage and for a few days after recovery. However, some people ('carriers') do not have symptoms may still carry the bacteria and be infectious, sometimes for months to years



Environmental factors for cholera outbreak

- ❑ Poor environmental sanitation
- ❑ Contaminated water and food ^{ex: sewage}
- ❑ Flies may carry vibrio ^{mechanical}
- ❑ Social factors:
 - ❑ Lack of education
 - ❑ poor quality of life
 - ❑ low standard personal hygiene
 - ❑ human habits



CLINICAL PICTURE

- The infection may be mild or asymptomatic in most of the cases, but sometimes it can be severe. Approximately 5–10% of persons suffer from severe cholera.
- Symptoms begin with sudden onset of **watery painless diarrhea**, which may be followed by vomiting
- The diarrhea has fishy odor in the beginning, but became less smelly & more watery over time.



CLINICAL PICTURE

- ❑ **Fever is typically absent.** *common in children*
- ❑ **Dehydration.** Dehydration can develop within hours after the onset of cholera symptoms. Depending on how many body fluids have been lost, dehydration can **range from mild to severe.**
- ❑ A loss of $\geq 10\%$ of total body weight indicates severe dehydration.



Clinical picture (continued)

- The classical textbook “**rice water**” **diarrhea**, which describes fluid stool with very little fecal material, appears within 24h from the start of the illness.
 - In severe cases stool volume exceeds 250 ml /kg leading to **severe dehydration, shock & death if untreated.**
- ... that also develop cholera gravis → cause profuse watery (accumulation of fluid in the gut) → no diarrhea but hypovolemic shock*



Cholera in children

- Breast feeding might get protection
- Symptoms are severe & fever is frequent.
- Shock, drowsiness & coma are common.
- Hypoglycemia is a recognized complication, which may lead to convulsions.
- Rotavirus infection may give similar picture & need to be excluded.

** regardless of cause of diarrhea correct management of dehydration and electrolyte imbalance can save life then find cause*



Lab diagnosis

- Organism can be seen in stool by direct microscopy after gram stain and dark field illumination is used to demonstrate motility.
- Cholera can be cultured on special alkaline media like triple sugar agar or Thiosulfate-citrate-bile salts-sucrose (TCBS) agar.

* Check cholera in:- suspected case

suspected outbreak

a percentage of diarrhoea → 1% in outbreak suspicion → in confirmed outbreak → 10% test needed every diarrhoea assume cholera

↳ in seasons beside winter



□ Serologic tests are available to define strains, but this is needed only during epidemics to trace the source of infection.



Treatment

- The primary goal of therapy is to **replenish fluid losses** caused by diarrhea & vomiting.
- Fluid therapy is accomplished in 2 phases:
rehydration and maintenance.
- Rehydration should be completed in **4 hours** & maintenance fluids should replace ongoing losses & provide daily requirement.



Fluid therapy

- ~~Ringer lactate~~ solution is preferred over normal saline because it corrects the associated metabolic acidosis.
ringier lactate
- IV fluids should be restricted to patients with severe dehydration.
- The oral route is preferred for maintenance & the use of ORS at a rate of 500-1000 ml/h is recommended. (up to 80% can be treated with ORS)

** if there's vomiting ORS could be difficult so maybe use IV*



Drug therapy

*not need necessarily since it's self-limiting
reverse the morbidic debilitation*

- The goals of drug therapy are to **eradicate infection**, **reduce morbidity** and **prevent complications**.
reduce symptomatic period
- The drugs used for adults include:
 - **Tetracycline or Doxycycline**
 - **Cotrimoxazole**
 - **Ciprofloxacin**



Drug therapy

- For children: **Erythromycin, Cotrimoxazole and Furazolidone** are the drugs of choice
- **Azithromycin** has been shown to be more effective than erythromycin and ciprofloxacin and is an appropriate first line regimen for **children and pregnant women**



Drug therapy

□ Zinc

□ *For children below 6 months of age, add **zinc** 10mg daily for 2 weeks.* } → even if improved

□ *For children from 6 months to 12 years, add zinc 20mg daily for 2 weeks*



Drug therapy

- Drug therapy reduces volume of stool & shortens period of hospitalization.
- It is only needed for few days (**3-5 days**).
- Drug resistance has been described in some areas & the choice of antibiotic should be guided by the local resistance patterns
- Antibiotic should be started when cholera is suspected **without waiting for lab confirmation.**

Complications

1- If dehydration is not corrected adequately & promptly it can lead to: **Hypovolemic shock** ,Acute renal failure & death.

2- Electrolyte imbalance is common.

3- Hypoglycemia occurs in children.

□ Complications of therapy like over hydration & side effects of drug therapy are rare *facial and eye puffiness*



proportion of Death from Cholera

- Most people who die, do so within the first day of symptoms appearing
- If untreated, ^{even no fluid} **50% of severe** ^{not all cases} cases are fatal
- With adequate re-hydration **less than 2%** will die
- With good surveillance, rapid establishment of re-hydration, and antibiotics for worst cases, almost all deaths can be avoided (**<0.2% die**)



Guideline for outbreak control

- Verification of diagnosis *(Contact)*
- Notification *→ CDC*
- Early case finding
- Establishment of treatment center *→ isolated hospital, cohort ward*
- Rehydration therapy
- Adjunct therapy
- Epidemiological investigations *→ finding source & exposure*
- Sanitation measures
- Chemoprophylaxis *(household contact), 300 mg doxycycline*
- Vaccination



Public health aspects *→ to prevent transmission*

- Isolation & barrier nursing is indicated
- Enteric precautions are always necessary
- Private room or special ward is required.
- All staff should wear a gown or apron and gloves when attending a patient or touching infective material.
- All **persons** should wash their hands on leaving the ward



Public health aspects

- ❑ Masks are not necessary but should be used if indicated.
- ❑ The discipline of the staff is of vital importance, especially in hand-washing, keeping the door closed, disinfection of toilets and bedpans etc
- ❑ Hands must be washed after touching the patient or contaminated articles, and contaminated articles should be discarded or bagged and labelled



Public health aspects

- Notification of the case to local authorities & WHO.
- Trace source of infection *→ propagated epidemic*
- Resume feeding with normal diet when vomiting has stopped & continue breastfeeding infants & young children.



Prevention

- ❖ Education on hygiene practices.
- ❖ Provision of safe, uncontaminated, drinking water to the people. *→ Basic human right*
- ❖ Antibiotic prophylaxis to house-hold contacts of index cases
- ❖ Vaccination against cholera to travelers to endemic countries & during public gatherings



Cholera vaccine

- Two new oral vaccines became available in 1997
- Both provoke a local immune response in the gut & a blood immune response.
- Cholera vaccination is no more required for international travelers because risk is small *→ just have good hygiene, don't drink untreated water*



Cholera Vaccines

2 types of Oral Cholera Vaccine:

2 types of Oral Cholera Vaccine:

A- Inactivated Oral Vaccines: Which are of types:

1- Dukoral (WC_rBS): monovalent inactivated vaccine; { two doses of 1 - 6 weeks apart },

→ all of the problem is this interval

□ it is administered with a buffer solution that, for adults, requires 150 ml of clean water. It is given to individuals above 2 years of age. ***It gives 2 Year protection***



Cholera Vaccines

2- Shanchol (BivWC) or Euvichol:

- It is a bivalent inactivated vaccine;
- They are given to all individuals over the **age of one year.**
- There must be a minimum of two weeks delay between each dose of these vaccines



Cholera Vaccines

(BivWC) or Euvichol: • - Shanchol (BivWC) or Euvichol:

- Two doses will provide protection against cholera for 3 years, while a single dose provides short term protection.
- Bacterial strains of both Inaba and Ogawa serotypes and of El Tor and Classical biotypes are included in the vaccine.



Cholera Vaccines

Live attenuated Oral Vaccine (Vaxchora). (FDA

2016): Vaxchora has been reported to reduce the chance of severe diarrhea in people by 90% at 10 days after vaccination and by 80% at 3 months after vaccination however the **safety and effectiveness** of Vaxchora in **pregnant or breastfeeding** women is **not yet known**



Cholera Vaccines

Live attenuated Oral Vaccine (Vaxchora)

Live attenuated

- It is also not known how long protection lasts beyond 3 – 6 months after getting the vaccine.
→ will not this good for residents
- Side effects from Vaxchora[®] are uncommon and may include tiredness, headache, abdominal pain, nausea and vomiting, lack of appetite, and diarrhea.



Cholera in Iraq

- Cholera outbreaks occur in Iraq about every 3 to 5 years and have a distinct seasonality that typically start in September and continue through December when cases naturally decline with a high probability of another outbreak the following spring

Cholera in Iraq

- **2007 outbreak:** First detected in Kirkuk on 14 August 2007. Cases have been reported by 11 governorates. 91% of all cases were reported from Sulaimani and Kirkuk provinces
- **2012 outbreak:** Number of laboratory confirmed Cholera in Sulaimani was 266 and Number of laboratory confirmed cases Acute watery Diarrhea cases 4693 *→ we had outbreak, every diarrhea is assumed cholera*

Cholera in Iraq *52*

- ❑ **2015 outbreak** : officially declared in September 2015, resulted in 4945 cases confirmed in 17 of the 18 governorates across Iraq
- ❑ **2022 outbreak** 865 confirmed cases of cholera, with four associated deaths. The most affected governorates are Kirkuk, Baghdad and Thi Qar
- ❑ **2023 outbreak** According to data from the Ministry of Health, 1,238 confirmed cholera cases including 7 deaths were registered in Iraq.
- ❑ Laboratory-confirmed cases of *Vibrio cholerae* 01 Inaba had been confirmed at the Central Public Health Laboratory in Baghdad



Cholera trend in Iraq

- Cholera outbreaks in Iraq seem to have a cyclic trend every 4–5 years.
- After the peak of 1998/1999, there were six peaks; in 2003, 2007, 2012, 2015; 2022 and 2023

Cholera in Iraq

